Solution to Quiz 6

Question 1

Is the polynomial $X^3 + Y^2X + Y(Y+1)$ irreducible in $\mathbb{C}[X,Y]$? Justify your answer.

Solution:

Thinking of this as a polynomial in the variable X with coefficients in $\mathbb{C}[Y]$, we see that it is a monic polynomial of degree 3. Thus, if it has a factor, then it has factor of the form $(X - \alpha)$ with α in the field of fractions of $\mathbb{C}[Y]$. Since $\mathbb{C}[Y]$ is a UFD and the polynomial is monic, this means that α is in $\mathbb{C}[Y]$.

In summary, if the polynomial has a factor, it has a root in $\mathbb{C}[Y]$.

A root must divide the "constant" coefficient in $\mathbb{C}[Y]$ which is $Y(Y+1)$. Thus, the possible values of roots are

$$\begin{aligned}
&X = c \text{ for } c \in \mathbb{C} - \{0\}\\
&X = cY \text{ for } c \in \mathbb{C} - \{0\}\\
&X = c(Y+1) \text{ for } c \in \mathbb{C} - \{0\}\\
&X = cY(Y+1) \text{ for } c \in \mathbb{C} - \{0\}
\end{aligned}$$

In each case, after substitution, we find that the result is a non-zero polynomial in $\mathbb{C}[Y]$ when $c \neq 0$:

$$\begin{aligned}
c^3 + c^2Y + Y(Y+1) &= Y^2 + (c^2+1)Y + c^3\\
c^3Y^3 + c^2Y^2 \cdot Y + Y(Y+1) &= (c^3+c^2)Y^3 + Y^2 + Y\\
c^3(Y+1)^3 + c^2(Y+1)^2 \cdot Y + Y(Y+1) &= \text{ terms of degree greater than 1 in } Y \ + c^3\\
c^3Y^3(Y+1)^3 + c^2Y^2(Y+1)^2 \cdot Y + Y(Y+1) &= c^3Y^6 + \text{ terms of degree less than 6 in } Y
\end{aligned}$$